Core Pure 1 · Chapter 9

Vectors

Lines and planes · Scalar products · Angles · Intersections · Distances

Chapter Overview

Lines and planes

1. Write vector and Cartesian equations of lines.
2. Describe a plane using two directions or a normal vector.
3. Use scalar products to find angles.

Geometry in three dimensions

4. Find intersections and distinguish skew lines.
5. Calculate shortest distances.
6. Reflect points and lines in planes.

Vector equation of a straight line in 3D 3

A vector equation describes every point on a straight line.

$$\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Find the points for $\lambda = 0, 1, 2, -1$.

Points

The first vector locates a fixed point. The second gives the direction.

Changing λ moves along the line in either direction.

The direction vector is non-zero.

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$$

Vector equation of a straight line in 3D 4

1. Find a line through $(2, 5, 1)$, parallel to direction $(1, 1, 0)$.

1

2. Find points on

$$\mathbf{r} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

at distance 3 from $(3, 4, 4)$.

2

Examples 5

1. Find a line through $(3, -5, 4)$, parallel to $(7, 0, -3)$.

1

2. Find a line through $A(4, 5, -1)$ and $B(6, 3, 2)$.

2

Use $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. Reversing the direction gives the same line.

Examples 6

1. The point $(a, b, 0)$ lies on

$$\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + t \begin{pmatrix} 1 \\ -6 \\ -2 \end{pmatrix}$$

Find a, b.

1

2. Show these equations describe the same line:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

2

Test Your Understanding So Far 7

Find a vector equation of the line through $A(3, 0, 2)$ and $B(-6, 1, 0)$.

Line

Examples 8

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

A circle with centre $P(2, 1, 3)$ meets this line at $A(0, -3, 6)$ and B.

a. Show that P is not on the line.

b. Find B.

a

b

Cartesian form of equation of straight line 9

For a line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$:

$x = a_1 + \lambda b_1,\ y = a_2 + \lambda b_2,\ z = a_3 + \lambda b_3$

Cartesian equation

This form assumes all three direction components are non-zero. If $b_2 = 0$, write $y = a_2$ separately; do not divide by zero.

A Cartesian equation has no free parameter.

Examples 10

1. Convert to Cartesian form.

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix}$$

1

2. Convert to Cartesian form.

$$\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

2

3. Convert $y = 3x + 2$ to vector form.

3

4. Convert $(x - 2)/3 = y + 5 = z/4$ to vector form.

4

5a. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ -9 \end{pmatrix}$$

Find p, q.

a

5b. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} -4 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -8 \end{pmatrix}$$

Find p, q.

b

5c. The point $(1, p, q)$ lies on

$$\mathbf{r} = \begin{pmatrix} 16 \\ -9 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Find p, q.

c

6. Show the lines are parallel:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

$$\frac{x-4}{2} = \frac{y+1}{-4} = \frac{z-3}{-8}$$

6

10. $P(2, 0, 4)$, $Q(a, 5, 1)$, $R(3, 10, b)$ are collinear. Find a, b.

10

11. On

$$\mathbf{r} = \begin{pmatrix} 8 \\ -5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -6 \end{pmatrix}$$

point A has $\lambda = -2$. Find the line through A, parallel to

$$\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$$

11

12. The point $A(4, a, 0)$ lies on

$$\mathbf{r} = \begin{pmatrix} 10 \\ 8 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ b \end{pmatrix}$$

a. Find a, b.

b. Find X, where $\lambda = -1$.

a

b

13. On

$$\mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

points A, B have parameters 5 and 2. Find AB.

13

14. On

$$\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

points C, A have parameters 4 and 3. A circle with centre C meets the line at A, B. Find $\overrightarrow{OB}$.

14

15. A circle has centre $(4, -1, 2)$ and radius $3\sqrt{5}$. It meets
$$x - 5 = \frac{y+1}{3} = \frac{z-6}{-2}$$
at two distinct points. Find them.

Hint: use $(5+\lambda, -1+3\lambda, 6-2\lambda)$ for a point on the line.

15

16. On
$$\mathbf{r} = \begin{pmatrix} -4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$
points A, B have parameters 2 and 5.

a. Find A, B.

b. Find the parallel line through $P(0, 2, 3)$.

c. Points C, D on this second line satisfy $AC = AD = AB$. Show P is their midpoint.

a

b

c

Equation of plane

19

A plane needs a fixed point and two non-parallel directions.

$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$

$\mathbf{a}$: a point on the plane.

$\mathbf{b}, \mathbf{c}$: independent directions within the plane.

$\lambda, \mu \in \mathbb{R}$: parameters.

For three non-collinear points A, B, C, use $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Examples

20

1. Find a plane through $A(2, 2, -1), B(3, 2, -1), C(4, 3, 5)$.

1

2. Show $P(2, 2, -1)$ lies on

$$\mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

2

Test Your Understanding 21

[June 2015 Q5] The plane contains $A(1,3,2)$, $B(-1,0,1)$, $C(2,1,0)$. Find its equation in the form $\mathbf{r}\cdot\mathbf{n}=p$.

Plane equation

Cartesian equation of a plane 22

A normal vector is perpendicular to a line (in 2D) or a plane (in 3D).

A line passes through $(0,3)$ and has normal $(-1,2)$. Find its equation.

Line

Cartesian equation of a plane 23

A plane passes through $(8, 4, -7)$ and has normal $(3, -2, 1)$. Find its equation.

Plane

Exercise 9B - Core Pure Year 1 Pages 177-178 24

3. Find k when the plane contains each point:

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

a. $(7, -1, k)$

a

b. $(1, k, 11)$

b

c. $(k, -4, 10)$

c

d. $(10, k, -k)$

d

7. Show $A(2,2,3)$, $B(1,5,3)$, $C(4,3,-1)$, $D(3,6,-1)$ are coplanar.

7

8. Show $A(2,3,4)$, $B(2,-1,3)$, $C(5,3,-2)$, $D(-1,-9,8)$ are not coplanar.

8

9. The plane is
$$\mathbf{r} = (3,-2,1) + \lambda(-2,3,5) + \mu(4,2,-3)$$
a. Find A at $\lambda = 1, \mu = 2$.
b. Show $B(1,-7,-1)$ is on the plane.
c. Find the line AB.
d. Find $C \neq A$ on this line with $|OC| = |OA|$.

a

b

c

d

Challenge

Show the line lies entirely in the plane:

$\Pi : \mathbf{r} = (2,3,0) + \lambda(1,-2,1) + \mu(2,-1,3)$

$l : \mathbf{r} = (2,6,1) + t(5,-7,6)$

Proof

Scalar Product 28

The scalar product is the sum of component products:

$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

It is a scalar, not a vector.

1. $(3,2,1) \cdot (4,0,1)$

1

2. $(5,-2,0) \cdot (3,3,1)$

2

3. $(a,5,1) \cdot (3,b,10)$

3

Angle between two vectors 29

For non-zero vectors:

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

For unit vectors, the scalar product is simply $\cos\theta$.

Find the angle between $(5, 3, 1)$ and $(1, 0, 5)$.

Angle

Important point... 30

Choose the correct directions

At B, the angle ABC uses $\overrightarrow{BA}$ and $\overrightarrow{BC}$: both point away from B.

Reversing one vector replaces θ by $180^\circ - \theta$.

Find the angle between $(2, 4, -1)$ and $(0, 1, 8)$.

Angle

Further Examples 31

For $A(2,3,5), B(5,0,4), C(4,-3,2)$:

a. Find $\angle ABC$.

b. Find the area of ABC.

a

b

Use the unrounded angle in the area calculation.

Perpendicular vectors 32

Non-zero vectors are perpendicular exactly when

$\mathbf{a} \cdot \mathbf{b} = 0$

1. Show $(2,3,1)$ and $(1,0,-2)$ are perpendicular.

2. Find a vector perpendicular to both $(-2,5,-4)$ and $(4,-8,5)$.

1

2

Angles between straight lines 33

$A = (2, 6, -1), B = (3, 4, 1)$. The line l_1 passes through A, B. The line l_2 passes through the origin parallel to $\mathbf{i} + \mathbf{k}$.

Find the acute angle between the lines.

a–b

c

Test Your Understanding 34

1. Show lines with directions $(2, 1, -1)$ and $(3, -1, 5)$ are perpendicular.

2. $ABCD$ is a parallelogram, with $A(2, -1, 5)$, $B(5, 2, 10)$, $D(-1, 1, 4)$.

a. Find $\overrightarrow{AB}$.

b. Give a vector equation of AB.

c. Find $\angle BAD$.

d. Find C.

e. Find the area.

1

2: a–b

2: angle

2: point and area

12. $A(7, 4, 4)$, $B(2, 2, 1)$. Find $\cos \angle AOB$ and the area of triangle OAB.

a

b

13. AB is a diameter of a circle centred at the origin. P is another point on the circumference. Prove $AP \perp BP$.

Proof

14. $A(5,-1,0)$, $B(2,4,10)$, $C(6,-1,4)$. Find CA, CB, the triangle area, all possible fourth vertices of a parallelogram, and its area.

a

b

c

d

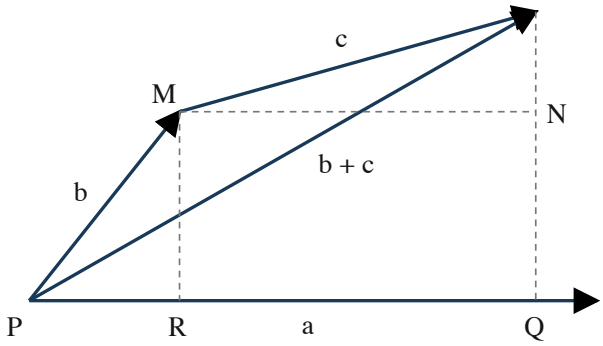
15. $P(1,-1,6)$, $Q(-2,5,4)$, $R(0,3,-5)$. Show PQR is right angled and find the centre and radius of its circumcircle.

Right angle

Circle

1. Use the definition of scalar product to prove commutativity.

1



2a. Show the three projection identities.

2b. Hence prove distributivity.

2a

2b

Vector Equations of Planes 40

A plane has normal $\mathbf{n}$, and contains a point with position vector $\mathbf{a}$.

For every point $\mathbf{r}$ on the plane, $\mathbf{r} - \mathbf{a}$ is perpendicular to $\mathbf{n}$.

Equation

In Cartesian form, if $\mathbf{n} = (a, b, c)$, the equation is $ax + by + cz = p$.

Example 41

Find the scalar-product and Cartesian equations of the plane through $(2, 3, -5)$ with normal $(3, 1, -1)$.

Plane

Angle between line and plane 42

Find the angle between

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ -12 \end{pmatrix}$$

and the plane $\mathbf{r} \cdot (2, -2, -1) = 2$.

The line–plane angle is complementary to the acute angle between the line and the normal.

Angle

Test Your Understanding 43

A plane has equation

$\mathbf{r} = (3,1,2) + \lambda(0,2,-1) + \mu(3,2,2)$

Show $(2,-1,-2)$ is normal to it. Find the acute angle between the plane and the line from $A(1,3,3)$ to $P(3,1,2)$.

a

b

Angle between two planes 44

Find the acute angle between the planes

$\mathbf{r} \cdot (4,4,-7) = 13$

$\mathbf{r} \cdot (7,-4,4) = 6$

Angle

Test Your Understanding 45

Find the acute angle between the planes

$\mathbf{r} \cdot (3, -1, 1) = 4$

$\mathbf{r} \cdot (2, 3, 0) = 7$

Angle

Exercise 9D - Core Pure Year 1 Pages 187-189 46

10. Show $A(3, 3, 7)$ lies on both lines and find their acute angle.

$$\frac{x-6}{-1} = \frac{y+3}{2} = \frac{z+2}{3}$$

$$\frac{x+5}{2} = \frac{y-15}{-3} = z - 3$$

a

b

11. $\mathbf{r} = \begin{pmatrix}1\\3\\3\end{pmatrix} + \lambda \begin{pmatrix}3\\2\\-1\end{pmatrix}$

$\mathbf{r} = \begin{pmatrix}3\\5\\-2\end{pmatrix} + \mu \begin{pmatrix}-4\\-3\\1\end{pmatrix}$

A is at $\lambda = 3$, B at $\mu = -2$. Find the acute angle between AB and the first line.

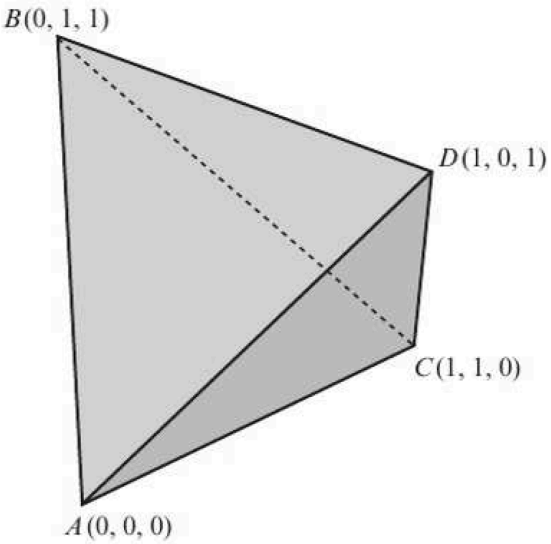
Angle

12. $A(3, 5, -1)$, $B(2, -2, 4)$, $C(4, 3, 0)$, $D(1, 4, -3)$. Show the points are not coplanar. Find the acute angle between AD and plane ABC.

a

b

13. A regular tetrahedron has vertices $A(0,0,0)$, $B(0,1,1)$, $C(1,1,0)$, $D(1,0,1)$. Show the angle between adjacent faces is $\cos^{-1}(1/3)$.



Proof

14. Ropes join the top $F(0,0,20)$ of a flagpole to $A(0,8,2)$, $B(12,-5,3)$, $C(-2,6,5)$. The pole is stable only if every pair of ropes makes an angle greater than $15°$. Is it stable?

Decision

Points of intersection 51

Show that these lines intersect and find the intersection:

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 1 \\ 4 \end{pmatrix}$$

Intersection

Skew lines are non-parallel lines that do not intersect. Always check all three components.

Test Your Understanding So Far 52

Prove the lines do not meet.

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Proof

Intersection of line and plane 53

Find the intersection of

$$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

and $\mathbf{r} \cdot (1, 2, 3) = 4$.

Intersection

Intersection of lines in Cartesian form 54

Prove these lines are skew:

$$\frac{x-2}{4} = \frac{y+3}{2} = z - 1$$

$$\frac{x+1}{5} = \frac{y}{4} = \frac{z-4}{-2}$$

Proof

2. Show these lines meet and find their intersection.

$$\mathbf{r} = \begin{pmatrix} -6 \\ 0 \\ 11 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ -2 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$$

Intersection

3. Show the lines do not meet.

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Proof

7. Find the intersection and the cosine of the acute angle between

$$\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

a

b

8. Prove these lines are skew.

$$\frac{x}{-3} = \frac{y+1}{5} = \frac{z-2}{4}$$

$$x = \frac{y-1}{-2} = \frac{z+5}{2}$$

Proof

11. Find where the line meets the plane and their acute angle.

$$\frac{x-3}{5} = \frac{y+2}{3} = 4 - z$$

$$4x + 3y - 2z = -10$$

a

b

Shortest distance between parallel lines 60

Find the shortest distance between the parallel lines

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$$

Distance

Shortest distance between any two lines 61

Find the shortest distance between

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

Distance

Shortest distance between point and line 62

Find the shortest distance from $A(1, 2, -1)$ to

$$\frac{x-1}{2} = \frac{y-1}{-2} = \frac{z+3}{-1}$$

and the perpendicular line through A.

Perpendicular

1. Find the shortest distance between the parallel lines

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$$

Distance

2. Find the shortest distance between

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$$

Distance

4. Find the shortest distance from $(4, 1, -1)$ to

$$\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

Distance

7. Reflect $P(3, 0, 2)$ in the line

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

Use the perpendicular foot as the midpoint of the point and its reflection.

Reflection

9. A birdwatcher is at $G(5, 4, 0.7)$ km and can spot birds within 500 m. A kestrel flies in a straight line from $A(3, 5, 0)$ to $B(12, 0, 1.2)$.

Can she spot it? Give a criticism of the model.

Path

a

b

11. Find the shortest distance from $A(3, -1, 2)$ to $\frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+1}{-2}$ and a Cartesian equation of the perpendicular through A.

a

b

Shortest Distance from Plane to Origin 69

For the plane $\mathbf{r} \cdot \mathbf{n} = k$, the shortest distance from the origin is

$$\boxed{d = \frac{|k|}{|\mathbf{n}|}}$$

The perpendicular meets the plane at $H = t\mathbf{n}$.

Derivation

The absolute value is essential: distance cannot be negative.

Shortest distance between point and plane 70

Find the shortest distance from $P(3, 2, -1)$ to $2x - 3y + z = 5$.

Distance

Use the normal $(2, -3, 1)$ and retain the sign of each coordinate in the numerator.

Test Your Understanding 71

Find the perpendicular distance from $(6, 2, 12)$ to the plane
$\mathbf{r} \cdot (3, -4, 2) = 5$

Distance

Reflections of points in planes 72

Find the distance of $P(1, 3, -2)$ from the plane $\mathbf{r} \cdot (1, 2, 2) = 5$. Then find the reflection Q of P in the plane.

Distance

Reflection

Reflections of lines in planes 73

Reflect the line

$$\frac{x-2}{2} = \frac{y-4}{-2} = z+6$$

in the plane $2x - 3y + z = 8$.

Fixed point

Reflected line

Exercise 9F (part 2) - Core Pure Year 1 Pages 191-192 74

6. Find the perpendicular distances to $10x + 10y + 23z = 81$. The normal has length 27.

a. From the origin.

b. From $(-1, -1, 4)$.

a

b

c. From $(2, 1, 3)$.

d. From $(6, 12, -9)$.

c

d

8. $P = (1, 0, 3)$, and $\Pi : -2x + y + z = 5$. Find the distance from P to Π, then its reflection Q.

a

b

10. $\Pi_1 : 3x - 2y + 4z = 6$. Find its distance from $(4, -1, 8)$.

For $\Pi_2 : \mathbf{r} = \lambda(2, -2, 3) + \mu(-3, 0, 3)$, show $(2, 5, 2)$ is normal and find the acute angle between the planes.

a

b

c

12. Reflect the line

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -3 \\ 4 \end{pmatrix}$$

in the plane $\mathbf{r} \cdot (1, -1, -1) = 4$.

Fixed point

Reflected line